

$$\begin{array}{c}
\mathcal{K}_{0^+}^{\#1} \quad \mathcal{K}_{0^+}^{\#2} \\
\mathcal{K}_{0^+}^{\#1} \dagger \begin{array}{|c|c|} \hline \frac{Y_1 k^2}{5} & -\frac{Y_1 k^2}{5} \\ \hline \end{array} \\
\mathcal{K}_{0^+}^{\#2} \dagger \begin{array}{|c|c|} \hline -\frac{Y_1 k^2}{5} & \frac{Y_1 k^2}{5} \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#1} \quad \mathcal{K}_{1^+}^{\#2} \\
\mathcal{K}_{1^+}^{\#1} \dagger^\alpha \begin{array}{|c|c|} \hline -\frac{Y_2 k^2}{15} & -\frac{Y_2 k^2}{3\sqrt{5}} \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#2} \dagger^\alpha \begin{array}{|c|c|} \hline -\frac{Y_2 k^2}{3\sqrt{5}} & -\frac{Y_2 k^2}{3} \\ \hline \end{array} \\
\mathcal{K}_{2^+}^{\#1} \quad \mathcal{K}_{2^+}^{\#2} \\
\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta} \begin{array}{|c|} \hline -\frac{4Y_1 k^2}{5} \\ \hline \end{array} \\
\mathcal{K}_{3^+}^{\#1} \quad \mathcal{K}_{3^+}^{\#2} \\
\mathcal{K}_{3^+}^{\#1} \dagger^{\alpha\beta\chi} \begin{array}{|c|} \hline -2Y_1 k^2 \\ \hline \end{array}
\end{array}$$

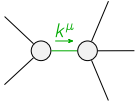
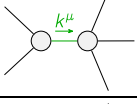
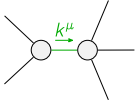
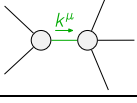
$$\begin{array}{c}
\mathcal{J}_{0^+}^{\#1} \quad \mathcal{J}_{0^+}^{\#2} \\
\mathcal{J}_{0^+}^{\#1} \dagger \begin{array}{|c|c|} \hline \frac{1}{2\text{Det}(0^+)} & -\frac{1}{2\text{Det}(0^+)} \\ \hline \end{array} \\
\mathcal{J}_{0^+}^{\#2} \dagger \begin{array}{|c|c|} \hline -\frac{1}{2\text{Det}(0^+)} & \frac{1}{2\text{Det}(0^+)} \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#1} \quad \mathcal{J}_{1^+}^{\#2} \\
\mathcal{J}_{1^+}^{\#1} \dagger^\alpha \begin{array}{|c|c|} \hline \frac{1}{6\text{Det}(1^+)} & \frac{\sqrt{5}}{6\text{Det}(1^+)} \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#2} \dagger^\alpha \begin{array}{|c|c|} \hline \frac{\sqrt{5}}{6\text{Det}(1^+)} & \frac{5}{6\text{Det}(1^+)} \\ \hline \end{array} \\
\mathcal{J}_{2^+}^{\#1} \quad \mathcal{J}_{2^+}^{\#2} \\
\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta} \begin{array}{|c|} \hline \frac{1}{\text{Det}(2^+)} \\ \hline \end{array} \\
\mathcal{J}_{3^+}^{\#1} \quad \mathcal{J}_{3^+}^{\#2} \\
\mathcal{J}_{3^+}^{\#1} \dagger^{\alpha\beta\chi} \begin{array}{|c|} \hline -\frac{1}{2k^2(\kappa_1 + \kappa_2)} \\ \hline \end{array}
\end{array}$$

Abbreviations used in matrices

$$\begin{aligned}
Y_1 &= \kappa_1^{(4)} + \kappa_2^{(4)} \& Y_2 = 6\kappa_1^{(4)} + \kappa_2^{(4)} \& \text{Det}(0^+) = \frac{2}{5}k^2(\kappa_1 + \kappa_2) \& \\
\text{Det}(1^+) &= -\frac{2}{5}k^2(6\kappa_1 + \kappa_2) \& \text{Det}(2^+) = -\frac{4}{5}k^2(\kappa_1 + \kappa_2)
\end{aligned}$$

Lagrangian

$$\begin{aligned}
&\kappa_1 \partial_\beta \mathcal{K}_\chi^\delta \partial^\alpha \mathcal{K}_\alpha^\beta + \kappa_2 \partial_\chi \mathcal{K}_\beta^\delta \partial^\alpha \mathcal{K}_\alpha^\beta + \frac{18}{5} \kappa_1 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \frac{18}{5} \kappa_2 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - \\
&\frac{12}{5} \kappa_1 \partial^\chi \mathcal{K}_\alpha^\beta \partial_\delta \mathcal{K}_{\beta\chi}^\delta - \frac{12}{5} \kappa_2 \partial^\chi \mathcal{K}_\alpha^\beta \partial_\delta \mathcal{K}_{\beta\chi}^\delta - 2 \kappa_1 \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} - 2 \kappa_2 \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi}
\end{aligned}$$

| Added source term(s):                                                              | $\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$ |                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source constraint(s)                                                               | # constraint(s)                                               | Covariant form                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                       |
| $\mathcal{J}_{0^+}^{\#2} + 3 \mathcal{J}_{0^+}^{\#1} = 0$                          | 1                                                             | $\partial_\beta \mathcal{J}^{\alpha\beta} = 0$                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                       |
| $\mathcal{J}_{1^+}^{\#2\alpha} = 5 \mathcal{J}_{1^+}^{\#1\alpha}$                  | 3                                                             | $6 \partial_\delta \partial_\chi \partial_\beta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\delta \partial_\chi \partial^\alpha \mathcal{J}^{\alpha\beta}_\beta = \partial_\delta \partial^\delta \partial_\chi \partial^\alpha \mathcal{J}^{\beta\chi}_\beta + 6 \partial_\delta \partial^\delta \partial_\chi \partial_\beta \mathcal{J}^{\alpha\beta\chi}$ |                                                                                                                                                                                                                       |
| Total # constraint(s):                                                             | 4                                                             |                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                       |
| Resolved pole(s)                                                                   | # polarization(s)                                             | Square mass                                                                                                                                                                                                                                                                                                                                                                                 | Residue                                                                                                                                                                                                               |
|  | 2                                                             | 0                                                                                                                                                                                                                                                                                                                                                                                           | $-\frac{1}{\kappa_1 + \kappa_2}$                                                                                                                                                                                      |
|  | 2                                                             | 0                                                                                                                                                                                                                                                                                                                                                                                           | $\frac{1}{\kappa_1 + \kappa_2}$                                                                                                                                                                                       |
|  | 2                                                             | 0                                                                                                                                                                                                                                                                                                                                                                                           | $\frac{80 \kappa_1^{(4)} + 55 \kappa_2^{(4)} - \sqrt{5} \sqrt{128 \kappa_1^{(4)2} + 416 \kappa_1^{(4)} \kappa_2^{(4)} + 413 \kappa_2^{(4)2}}}{6 \kappa_1^{(4)2} + 7 \kappa_1^{(4)} \kappa_2^{(4)} + \kappa_2^{(4)2}}$ |
|  | 2                                                             | 0                                                                                                                                                                                                                                                                                                                                                                                           | $\frac{80 \kappa_1^{(4)} + 55 \kappa_2^{(4)} + \sqrt{5} \sqrt{128 \kappa_1^{(4)2} + 416 \kappa_1^{(4)} \kappa_2^{(4)} + 413 \kappa_2^{(4)2}}}{6 \kappa_1^{(4)2} + 7 \kappa_1^{(4)} \kappa_2^{(4)} + \kappa_2^{(4)2}}$ |
| Resolved unitarity condition(s):                                                   |                                                               | (Demonstrably impossible)                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                       |